Yijie Pan

Ph.D. Candidate in Mathematics, University of Southern California

338C Kaprielian Hall, Los Angeles, CA · yijiepan@usc.edu · ypan-code.github.io

RESEARCH INTERESTS

Symplectic and contact topology, Reeb dynamics, symplectic capacities, and gauge theory.

EDUCATION

2024–present **University of Southern California**, Ph.D. in Mathematics
Advisor: Julian Chaidez

Dec. 2023 **University of California, Los Angeles**, B.S. in Mathematics

PAPERS AND PREPRINTS

1. *Pseudo-Anosov Reeb flows and contact structures*, with Julian Chaidez. Preprint, 2025. arXiv:2512.12698.
2. *Polytopal counterexamples to the equality case of Viterbo's conjecture*, with Osagie Osayande and Henry Riley. Preprint, 2026.

WORK IN PROGRESS

1. *On the gluing formula of the Bauer–Furuta invariant*, with Baran Çetin.
2. *Rigidity of Besse contact forms on the 5-sphere*, with Baran Çetin and Michael Huang.
3. *Symplectic TQFT for stable Hamiltonian structures*, with Michael Huang.

TALKS

Oct. 2026 *Pseudo-Anosov Reeb flows and contact structures*
AMS Fall Eastern Sectional Meeting, Special Session on Recent Developments at the Interface of Hyperbolic Dynamics and Geometric Structures, George Washington University, Washington, DC

Sep. 2026 *Crash course in analysis: pseudo-holomorphic curves and somewhere injectivity*
USC Topology Seminar on Symplectic Field Theory

Nov. 2025 *The classical Bauer–Furuta invariant*
Gauge Theory Seminar, University of Southern California

Jan. 2024 *Contact manifolds, I*
USC Student Topology Seminar

SERVICE

Fall 2026 Co-organizer, USC Topology Seminar on Symplectic Field Theory (with Louis-Simon Cyr and Michael Huang)

Summer 2026	Graduate mentor, Research Experience for Undergraduates, University of Southern California Mentored the project that led to "Polytopal counterexamples to the equality case of Viterbo's conjecture."
2024–2025	Graduate mentor, Directed Reading Program, University of Southern California Fall 2024: knot theory and knot invariants. Spring 2025: algebraic topology and Adams's Hopf invariant one theorem. Fall 2025: symplectic reduction and geometric invariant theory (with Baran Çetin).

EXPOSITORY NOTES

Sep. 2026	*Pseudo-holomorphic curves and somewhere injectivity* — talk notes, USC Topology Seminar on Symplectic Field Theory
Nov. 2025	*The classical Bauer–Furuta invariant* — talk notes, USC Gauge Theory Seminar
Dec. 2024	*Representations of* $\mathfrak{sp}_4(\mathbb{C})$
Nov. 2023	*Atiyah–Hirzebruch's counterexample to the integral Hodge conjecture*
Nov. 2023	*K-theory and the Brouwer fixed point theorem*

Last updated October 2026